

Arrangements & Conditionalities and Groupings 1

SET 1: (CAT 2019-Slot-1)

Refer to the data below and answer the questions that follow.

Princess, Queen, Rani and Samragini were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancers. The dancers performed their second items in the same sequence of their performance of their first items.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen
- No composer who assigned item to Rani, assigned any item to Samragini
- The first performance was by Princess; this item was assigned by Badal
- The last performance was by Rani; this item was assigned by Gagan
- The items assigned by Ashman were performed consecutively
- The number of performances between items assigned by each of the remaining composers was the same.

Question 1

Which of the following is true?

- 1) The second performance was composed by Gagan.
- 2) The third performance was composed by Ashman.
- 3) The second performance was composed by Dyu.
- 4) The third performance was composed by Dyu.

Question 2

Which of the following is FALSE?

- 1) Rani did not perform in any item composed by Badal.
- 2) Princess did not perform in any item composed by Dyu.
- 3) Samragini did not perform in any item composed by Ashman.
- 4) Queen did not perform in any item composed by Gagan.

Question 3

The sixth performance was composed by:

- 1) Dyu
- 2) Gagan
- 3) Ashman
- 4) Badal

Question 4

Which pair of performances were composed by the same composer?

- 1) The second and the sixth
- 2) The third and the seventh
- 3) The first and the seventh
- 4) The first and the sixth



SET 2: (CAT 2017-Slot-2)

Refer to the data below and answer the questions that follow.

A tea taster was assigned to rate teas from six different locations - Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 10. He gave a unique integer rating to each tea. Some other information is given below:

1. Cup 6 contained tea from Himachal.
2. Tea from Ooty got the highest rating, but it was not in Cup 3.
3. The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
4. Only two cups got ratings in even numbers.
5. Cup 2 got the minimum rating and this rating was an even number.
6. Tea in Cup 3 got a higher rating than that in Cup 1.
7. The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Question 1

What was the second highest rating given?

Question 2

What was the number of the cup that contained tea from Ooty?

Question 3

If the tea from Munnar did not get the minimum rating, what was the rating of the tea from Wayanad?

- 1) 3 2) 5 3) 1 4) 6

Question 4

If cups containing teas from Wayanad and Ooty had consecutive numbers, which of the following statements may be true?

- 1) Cup 5 contains tea from Assam
- 2) Cup 1 contains tea from Darjeeling
- 3) Tea from Wayanad has got a rating of 6
- 4) Tea from Darjeeling got the minimum rating



SET 3: (CAT 2020-Slot-3)

Refer to the data below and answer the questions that follow.

Sixteen patients in a hospital must undergo a blood test for a disease. It is known that exactly one of them has the disease. The hospital has only eight testing kits and has decided to pool blood samples of patients into eight vials for the tests. The patients are numbered 1 through 16, and the vials are labelled A, B, C, D, E, F, G, and H. The following table shows the vials into which each patient's blood sample is distributed.

Patient	Vials	Patient	Vials
1	B, D, F, H	9	A, D, F, H
2	B, D, F, G	10	A, D, F, G
3	B, D, E, H	11	A, D, E, H
4	B, D, E, G	12	A, D, E, C
5	B, C, F, H	13	A, C, F, H
6	B, C, F, G	14	A, C, F, G
7	B, C, E, H	15	A, C, E, H
8	B, C, E, G	16	A, C, E, G

If a patient has the disease, then each vial containing his/her blood sample will test positive. If a vial tests positive, one of the patients whose blood samples were mixed in the vial has the disease. If a vial tests negative, then none of the patients whose blood samples were mixed in the vial has the disease.

Question 1

Suppose vial C tests positive and vials A, E and H test negative. Which patient has the disease?

- 1) Patient 6 2) Patient 14 3) Patient 8 4) Patient 2

Question 2

Suppose vial A tests positive and vials D and G test negative. Which of the following vials should we test next to identify the patient with the disease?

- 1) Vial E 2) Vial C 3) Vial H 4) Vial B

Question 3

Which of the following combinations of test results is NOT possible?

- 1) Vials A and E positive, vials C and D negative
2) Vial B positive, vials C, F and H negative
3) Vials A and C positive, vials D and E negative
4) Vials B and D positive, vials F and H negative

Question 4

Suppose one of the lab assistants accidentally mixed two patients' blood samples before they were distributed to the vials. Which of the following correctly represents the set of all possible numbers of positive test results out of the eight vials?

- 1) {4,5,6,7,8} 2) {4,5} 3) {4,5,6,7} 4) {5,6,7,8}



SET 4: (CAT 2021-Slot-1)

Refer to the data below and answer the questions that follow.

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four of the authors were from India and two each were from Japan and China. Each author belonged to exactly one of the three areas — Manufacturing, Automation and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

The following facts are also known.

1. F, an Indian author from the Logistics area, wrote only one paper. It was scheduled in the October issue.
2. A was from the Automation area and did not have a paper scheduled in the October issue.
3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.
4. A and H were from different countries, but had their papers scheduled in exactly the same issues.
5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.
6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.
8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
9. C and H belonged to different areas.

Question 1

What is the correct sequence of number of papers written by B, C, E and G, respectively?

- 1) 1, 2, 2, 3 2) 3, 1, 1, 3 3) 1, 2, 2, 1 4) 1, 3, 3, 1

Question 2

How many papers were written by Indian authors?

Question 3

Which of the following statement(s) MUST be true?

Statement A: Every issue had at least one paper by author(s) from each country.

Statement B: Every issue had at most two papers by author(s) from each area.

- 1) Only Statement B 2) Both the statements
3) Neither of the statements 4) Only Statement A

Question 4

Which of the following statements is FALSE?

- 1) Every issue had exactly two papers by authors from Logistics area.
- 2) Every issue had exactly one paper by a Chinese author.
- 3) Every issue had exactly two papers by Indian authors.
- 4) Every issue had at least one paper by author(s) from Automation area.

